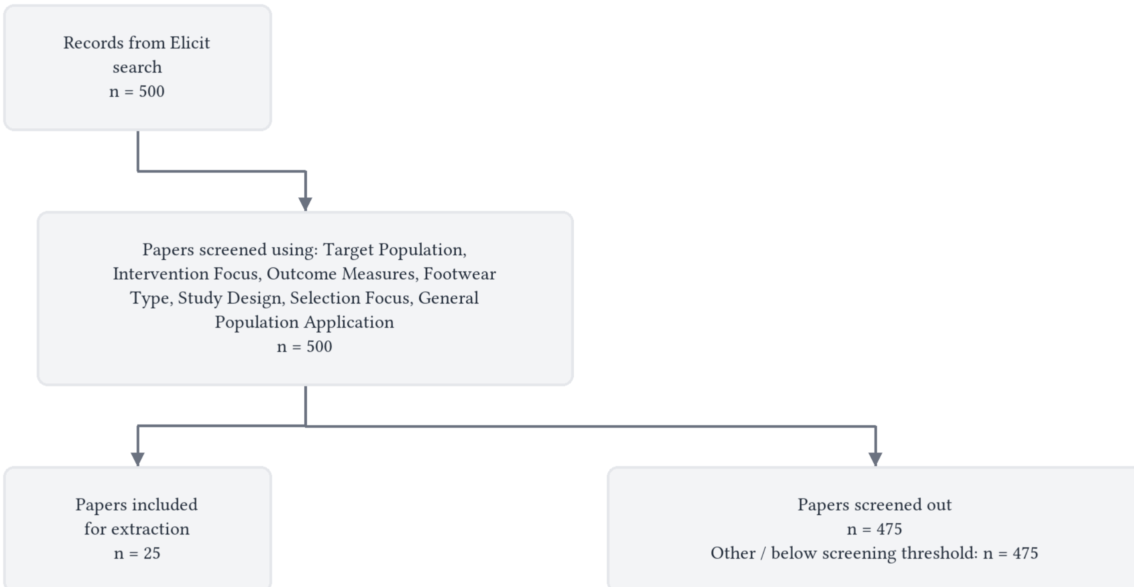
sports shoe selection guide

Prioritize lightweight, comfortable shoes that feel right to you over traditional arch-based selection methods, with attention to sport-specific needs like traction in court and field sports.

Abstract

Large-scale randomized controlled trials totaling over 7,200 military recruits found that selecting running shoes based on foot arch height or plantar shape does not reduce injury risk (rate ratio 0.97 for both men and women, 95% CI: 0.88-1.06 and 0.85-1.08 respectively) [1]. A Cochrane systematic review of 11,240 participants confirmed that prescribing footwear based on foot posture makes little or no difference to lower-limb running injuries (rate ratio 1.03, 95% CI 0.94-1.13) [2], and comparisons between neutral/cushioned, minimalist, motion control, and stability shoes showed similarly null effects on injury rates [2]. However, specific shoe features demonstrate benefits in targeted contexts: lighter shoes with lower heel-to-toe drop and wider toe boxes reduced bone stress injuries by 44% in male military trainees [3], shoe mass consistently affects running economy (approximately 1% increase in oxygen consumption per 100 grams added) [4], and basketball performance depends primarily on outsole traction ($p = 0.005$) and forefoot bending stiffness ($p = 0.013$) rather than weight [5]. Individual responses to shoe features vary by arch height (low-arched runners showed lower loading rates in motion control shoes while high-arched runners benefited from cushion trainers) [6] and experience level (low drop shoes may reduce injuries for occasional but not regular runners) [7]. Given the lack of evidence supporting traditional prescription paradigms and the strong individual comfort preferences that can be manipulated independently of biomechanical effects [8], current evidence supports prioritizing lightweight, comfortable shoes without assuming pronation control features are necessary for injury prevention [4], while recognizing that sport-specific features like traction matter more than universal characteristics for certain activities [5].

Flow Diagram



Paper search

We performed a semantic search across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We ran this query: "sports shoe selection guide"

The search returned 500 total results from Elicit.

We retrieved 500 papers most relevant to the query for screening.

Screening

We screened in sources based on their abstracts that met these criteria:

- **Target Population**: Does the study involve participants who engage in sports or physical activities requiring specialized footwear?
- **Intervention Focus**: Does the study examine sports shoe selection methods, criteria, or guidelines?
- **Outcome Measures**: Does the study evaluate the impact of different shoe selection approaches on performance, injury rates, comfort, or biomechanical parameters?
- **Footwear Type**: Does the study focus on athletic footwear, sports shoes, or sport-specific footwear (rather than casual, dress, or non-athletic footwear)?
- **Study Design**: Is the study an experimental study (randomized controlled trial, quasi-experimental), observational study (cohort, cross-sectional, case-control), systematic review, or meta-analysis?

- **Selection Focus**: Does the study address selection criteria rather than focusing solely on shoe design, manufacturing, or material properties without addressing selection?
- **General Population Application**: Does the study address general sports shoe selection rather than focusing exclusively on foot pathology treatment or orthotic interventions?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

At abstract screening, the number of papers excluded for each primary reason was:

- **Other / below screening threshold**: n = 475

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Sport and Population**:

  Extract the specific sport(s) and target population studied for sports shoe selection guidance, including:

- Sport type (running, football/soccer, basketball, etc.)
- Athlete level (amateur, recreational, professional, youth, etc.)
- Population characteristics (age range, gender, sample size)
- Any specific subgroups analyzed (e.g., injury-prone athletes, different skill levels)

- **Shoe Features Studied**:

  Extract all specific shoe characteristics or features investigated for selection guidance, including:

- Structural features (cushioning level, heel-to-toe drop, midsole stiffness, outsole design, upper materials)
- Sizing and fit aspects (width, length, 3D shape matching, custom fitting)
- Performance-related features (weight, flexibility, traction elements)
- Any other technical specifications studied Note whether features were studied individually or in combination.

- **Selection Methods**:

  Extract the shoe selection approaches, criteria, or systems studied, including:

- Traditional methods (length/width measurement, trying on shoes)
- Advanced fitting systems (3D foot scanning, pressure mapping, gait analysis)
- Professional guidance vs. self-selection
- Specific selection criteria used (comfort, brand preference, price, recommendations)
- Any systematic selection protocols or decision-making frameworks tested

- **Individual Factors**:

  Extract personal characteristics that were studied as factors influencing sports shoe selection, including:

- Foot type/morphology (arch height, foot shape, biomechanical patterns)
- Injury history or risk factors
- Running/playing style or technique
- Training volume, intensity, or experience level
- Personal preferences or comfort requirements
- Any other individual differences that affected shoe selection recommendations

- **Performance Outcomes**:

  Extract all performance-related results from different shoe selection approaches or shoe features, including:

- Biomechanical measures (gait parameters, force production, efficiency)
- Athletic performance metrics (speed, agility, ball control, accuracy)
- Comfort and preference ratings
- Any objective performance tests conducted Include effect sizes, significance levels, and direction of effects where reported.

- **Injury Outcomes**:

  Extract all injury-related findings from sports shoe selection studies, including:

- Injury rates or risk (overall and by body region)
- Specific injury types prevented or caused
- Relationship between shoe features and injury risk
- Injury recurrence rates
- Time to injury or injury-free periods Include statistical measures, odds ratios, confidence intervals where available.

- **Selection Recommendations**:

  Extract specific guidance, recommendations, or conclusions about sports shoe selection provided by the study, including:

- Best practices for shoe selection process
- Optimal shoe features for different types of athletes or sports
- Personalization strategies (how to match shoes to individual needs)
- What factors to prioritize when selecting shoes
- Any decision-making frameworks or algorithms proposed
- Warnings about poor selection practices to avoid

- **Study Limitations**:

  Extract key limitations and considerations that affect the applicability of shoe selection guidance, including:

- Sample size or population limitations
- Study duration and follow-up periods
- Measurement limitations or validity concerns
- Generalizability issues (specific to certain sports, populations, or conditions)
- Conflicts of interest (industry funding, author affiliations)
- Any factors that limit the strength or applicability of the selection recommendations

Results

Characteristics of Included Studies

The review included 25 sources examining sports shoe selection across multiple sports and populations. The majority focused on running (21 sources), with smaller numbers addressing basketball [5], American football [9], soccer/football [10], and field hockey [11].

Study	Full Text Retrieved?	Sport	Population	Sample Size	Study Design
J. Knapik et al., 2009 [12]	Yes	Running	Military recruits (amateur)	3,062 total (E: 1,530; C: 1,532)	RCT
Carolina Romero Cardenas, 2024 [13]	Yes	Running	Amateurs, novices, beginners	Not specified	Review
Shuangshuang Lin et al., 2022 [7]	Yes	Running	General runners	41 articles reviewed	Systematic review
Fabian Hoitz et al., 2020 [14]	No	Running	Novice, recreational, high calibre	68 articles reviewed	Systematic review
C. Richards et al., 2008 [15]	No	Running	Adult recreational/competitive	Not applicable	Systematic review
J. Wannop et al., 2018 [9]	Yes	American football	Professional (NFL)	1,451	Descriptive review
Rodrigo Éberte Andrade & T. R. Santos, 2022 [16]	No	Running	Amateur	254	Observational study
Daniel Joachim Kulessa et al., 2017 [10]	No	Football/Soccer	Not specified	~100 publications	Review
R. Butler et al., 2006 [6]	No	Running	Recreational (>10 mi/wk)	40 (20 high-arch, 20 low-arch)	Controlled laboratory study
C. Agresta et al., 2022 [4]	Yes	Running	Recreational	Not specified	Narrative review
Santiago A. Ruiz-Alias et al., 2022 [17]	No	Running	Trained runners	36 articles, 61 shoe models	Systematic review
J. Worobets & J. Wannop, 2015 [5]	No	Basketball	Recreational	20	Experimental study
J. Knapik et al., 2014 [1]	No	Running	Military recruits	7,203 total (Army: 3,119; Air Force: 2,673; Marines: 1,411)	Meta-analysis of RCTs
Patrick Mai et al., 2023 [18]	Yes	Running	Adults (primarily male)	107 articles	Scoping review
C. Napier & R. Willy, 2018 [19]	Yes	Running	Experienced runners	Not applicable	Commentary
Iago Morais Leão et al., 2025 [20]	No	Running	Not specified	Not specified	Integrative review

Study	Full Text Retrieved?	Sport	Population	Sample Size	Study Design
N. Relph et al., 2022 [2]	No	Running	Adult runners, military personnel	11,240 (12 trials)	Cochrane systematic review
R. Tedeschi, 2024 [21]	No	Running	Adult runners (various levels)	4 articles	Scoping review
Korey B Kasper et al., 2023 [3]	No	Physical training	Military trainees (BMT)	Not specified	RCT
M. Schwellnus, 2011 [22]	Yes	Running	Marine Corps recruits	Not specified	Evidence-based review
J. Knapik et al., 2009a [23]	No	Running	Military recruits (BCT)	3,062 total (E: 1,530; C: 1,068)	RCT
F. I. Kleindienst et al., 2006 [24]	No	Running	Not specified	471 (244 female, 227 male)	Field study
F. Kleindienst et al., 2006 [25]	No	Running	Not specified	471 (244 female, 227 male)	Field study
A. Fife et al., 2025 [8]	No	Running	Female runners	21	Single-blind crossover RCT
Christopher R Derry et al., 2024 [11]	No	Field hockey	Not specified	401	Online survey

Most studies examined running shoe selection, with sample sizes ranging from 20 participants [5] to over 11,000 across multiple trials [2]. Study designs included randomized controlled trials, observational studies, systematic reviews, and surveys. Full text was available for approximately one-third of sources, with the remainder available as abstracts only.

Shoe Features and Selection Methods Investigated

Structural and Performance Features

Studies examined a wide range of shoe characteristics that could inform selection decisions. The most commonly investigated structural features included cushioning level [4, 7, 13], heel-to-toe drop [3, 7, 13, 21], and midsole properties [4, 7, 18]. Specific midsole parameters studied included thickness (optimal range 15-20 mm) [7], hardness (Asker C50-C55) [7], and stiffness characteristics [4, 13, 18].

Feature Category	Specific Features Studied	Sources
Midsole characteristics	Thickness (15-20 mm) [7], hardness (C50-C55) [7], material (EVA, PU) [13], stiffness [4, 13]	Lin et al., 2022; Cardenas, 2024; Agresta et al., 2022
Heel-toe drop	Low drop [21], medium drop [13], high drop [13], 10mm drop [16]	Tedeschi, 2024; Cardenas, 2024; Andrade & Santos, 2022

Feature Category	Specific Features Studied	Sources
Outsole design	Traction elements [5, 18], stud design [11], heel flare (15°) [7], outsole profile [18]	Worobets & Wannop, 2015; Mai et al., 2023; Derry et al., 2024; Lin et al., 2022
Upper materials	Lacing systems [13], stiffness [13], elasticity [14]	Cardenas, 2024; Hoitz et al., 2020
Performance features	Weight/mass [3, 5, 7, 13], flexibility [13, 18], bending stiffness [5, 7]	Multiple sources
Advanced features	Carbon plates [7, 13], 3D printed heel cups [7], TPU materials [13]	Cardenas, 2024; Lin et al., 2022
Support systems	Motion control [1, 12], stability features [1, 12], pronation control [15]	Knapik et al., 2009; Knapik et al., 2014; Richards et al., 2008
Fit and sizing	3D shape matching [9], width options [11], toe box width [3]	Wannop et al., 2018; Derry et al., 2024; Kasper et al., 2023

For basketball, the key features studied were outsole traction, forefoot bending stiffness, and shoe mass [5]. In football/soccer, outsole configuration and its influence on traction were most studied [10], along with shoe construction's effect on ball interactions [10]. Field hockey players prioritized fit, comfort, support, cushioning, and stud design [11].

Selection Methods and Approaches

Traditional foot-based selection methods dominated the literature, particularly the practice of matching shoes to foot arch height or plantar shape. This approach typically involved assigning motion control shoes to low arches, stability shoes to medium arches, and cushioned shoes to high arches [1, 12, 23].

Advanced fitting systems included 3D foot scanning technology [9], with the NFL implementing an objective measurement system to match the 3D shape of athletes' feet to internal shoe shapes [9]. Gait analysis was used to inform shoe recommendations [8], though evidence for its effectiveness was mixed.

Professional guidance versus self-selection varied across studies. Most amateur runners in one survey did not receive guidance for shoe selection [16], while others emphasized the importance of recommendations based on individual runner needs [13]. Specific selection criteria identified by runners included comfort [11, 16], intermediate cushioning [16], and heel-to-toe drop of approximately 10mm [16].

Effects on Performance and Injury

Performance Outcomes

Shoe weight demonstrated consistent effects on performance across sports. In running, every 100 grams of added shoe mass increased oxygen consumption by approximately 1% [4]. Elite runners using shoes with carbon plates showed performance improvements of 4-6% [13], though the exact mechanisms remained debated [4].

For basketball, outsole traction had the largest influence on performance. When traction decreased by 20%, participants performed significantly worse in all tests ($p < 0.001$) [5]. Conversely, when traction increased by 20%, performance improved significantly in cutting drills ($p = 0.005$) [5]. Forefoot bending stiffness had moderate effects on sprint and cutting performance (p = 0.013 and p = 0.016 respectively) [5], while shoe mass showed no effect on basketball performance [5].

Performance Measure	Shoe Feature	Effect	Statistical Significance	Source
Oxygen consumption	100g mass increase	~1% increase	Not specified	Agresta et al., 2022 [4]
Running time	Carbon fiber plates	4-6% improvement	Not specified	Cardenas, 2024 [13]
Basketball cutting	20% traction increase	Better performance	p = 0.005	Worobets & Wannop, 2015 [5]
Basketball sprint	Forefoot stiffness	Moderate improvement	p = 0.013	Worobets & Wannop, 2015 [5]
Loading rate	Motion control (low arch)	Lower rate	Significant interaction	Butler et al., 2006 [6]
Loading rate	Cushion trainer (high arch)	Lower rate	Significant interaction	Butler et al., 2006 [6]
Rearfoot motion	Motion control shoes	Better control	Significant main effect	Butler et al., 2006 [6]
Shock attenuation	Cushion trainer shoes	Better attenuation	Significant main effect	Butler et al., 2006 [6]

Biomechanical measures varied by shoe type and individual characteristics. Low-arched runners showed lower instantaneous loading rates in motion control shoes, while high-arched runners had lower rates in cushion trainer shoes [6]. Motion control shoes controlled rearfoot motion better than cushion trainers, while cushion trainers provided superior shock attenuation [6].

Midsole characteristics influenced running mechanics. Optimal thickness of 15-20 mm and hardness of Asker C50-C55 were beneficial for performance [7]. Increased bending stiffness reduced energy loss and improved running economy [7], though effects varied by individual runner characteristics. Heel flare reduced rearfoot movement [7], and heel stabilizers improved rearfoot stability while reducing musculoskeletal transients [7].

Injury Outcomes

The most robust evidence came from large-scale randomized controlled trials examining traditional arch-based shoe prescription. A meta-analysis of three military studies totaling 7,203 participants found little difference in injury rates between experimental groups (shoes matched to arch height) and control groups (standard stability shoes) [1]. For men, the summary rate ratio was 0.97 (95% CI: 0.88, 1.06) [1], and for women it was 0.97 (95% CI: 0.85, 1.08) [1].

Individual studies confirmed these findings. In US Army Basic Combat Training with 3,062 participants, selecting shoes based on plantar shape showed minimal influence on injury risk for men (RR 1.01, 95% CI 0.88-1.16, p = 0.87) and women (RR 1.07, 95% CI 0.91-1.25, p = 0.44) [12].

Shoe Comparison	Population	Injury Risk Ratio	95% CI	Evidence Quality	Source
Neutral/cushioned vs. minimalist	Adult runners	0.77	0.59 to 1.01	Low	Relph et al., 2022 [2]
Motion control vs. neutral/cushioned	Adult runners	0.92	0.30 to 2.81	Very low	Relph et al., 2022 [2]
Soft vs. hard midsole	Adult runners	0.82	0.61 to 1.10	Low	Relph et al., 2022 [2]
Stability vs. neutral/cushioned	Adult runners	0.49	0.18 to 1.31	Very low	Relph et al., 2022 [2]
Motion control vs. stability	Adult runners	3.47	1.43 to 8.40	Very low	Relph et al., 2022 [2]
Foot posture-based prescription	Military recruits	1.03	0.94 to 1.13	Moderate	Relph et al., 2022 [2]
Arch-based prescription (men)	Military recruits	0.97	0.88 to 1.06	Meta-analysis	Knapik et al., 2014 [1]
Arch-based prescription (women)	Military recruits	0.97	0.85 to 1.08	Meta-analysis	Knapik et al., 2014 [1]

Specific shoe features showed more promising injury prevention potential. In US Air Force basic military training, lighter-weight shoes with lower heel-to-toe drop and wider toe box reduced bone stress injuries among male trainees by 43.62% (absolute risk reduction 6.05%, from 13.87% to 7.82%) [3]. Shoes with thinner midsoles reduced injury risk at the knee joint [14].

Heel-toe drop effects on injury varied by runner experience. Low drop shoes may reduce injury rates for occasional runners but increase risk for regular runners [7]. Medium-low drop shoes were suggested for preventing patellofemoral pain ($p = 0.003$) [13].

Specific injury types addressed in the literature included stress fractures [13], shin splints [13], patellar tendonitis [13], plantar fasciitis [13], and Achilles tendonitis [13]. Motion-controlled shoes were proposed to help prevent pronation-related injuries [20], though evidence quality was limited. Heel flare reduced rearfoot movement and may mitigate ankle joint injuries [7]. Personalized heel cups showed benefits for plantar heel pain [7].

One notable exception to the generally null findings was a study showing motion control shoes protected against injury in experienced runners with pronated feet [19], though methodological differences across studies complicated interpretation.

Comfort and Preference

Subjective perceptions of shoes showed significant variation based on shoe description and context. In a single-blind crossover trial with 21 female runners, own shoes received the highest comfort ratings (83.3 ± 3.8 mm on a 100mm scale), followed by shoes labeled as "gait-matched" (66.1 ± 21.5 mm) and "basic" shoes (49.0 ± 24.1 mm) [8]. The same experimental shoes received different ratings depending on how they were described, with gait-matched shoes scoring 15.6 mm higher on comfort (95% CI: 5.7, 25.5) [8], 17.1 mm higher on performance (95% CI: 5.6, 28.6) [8], and 30.1 mm higher on injury reduction (95% CI: 8.9, 51.2) [8] compared to basic shoes [8].

Most runners preferred their own shoes (71.4%), followed by gait-matched shoes (23.8%) and basic shoes (4.8%) [8]. However, spatiotemporal and kinematic parameters showed no significant differences between shoes ($p \geq 0.157$) [8], indicating that subjective perceptions can be manipulated independently of biomechanical changes.

Comfort also showed associations with running economy in some studies, with more comfortable shoes improving economy by 0.7% [4], though other studies found no significant differences [4].

Synthesis of Findings

The literature reveals an apparent paradox: traditional arch-based shoe prescription methods show consistently null effects on injury risk despite being widely recommended, while certain specific shoe features demonstrate injury reduction potential in targeted contexts. This heterogeneity requires explanation beyond simply counting positive versus negative studies.

Methodological Quality and Population Context

The strongest evidence against arch-based prescription comes from large, well-controlled military studies. The meta-analysis of 7,203 military recruits across three branches used rigorous randomization, objective injury assessment through medical records, and controlled for known confounders [1]. These studies consistently found rate ratios near 1.0 with narrow confidence intervals [1], suggesting true null effects rather than underpowered studies.

In contrast, studies showing benefits of specific shoe features often examined different populations and outcomes. The 43.62% reduction in bone stress injuries from lighter shoes [3] occurred in a high-intensity military training context where bone stress is a predominant injury mechanism. The population (male military trainees undergoing standardized physical training) differs substantially from recreational runners with varied training patterns, potentially explaining why this benefit wasn't observed in broader populations.

The evidence quality hierarchy also matters. Most evidence comparing different shoe types was rated low to very low certainty due to inability to blind participants, variation in injury definitions, and limited sample sizes [2]. The moderate-certainty evidence that foot posture-based prescription makes little difference (Rate Ratio 1.03, 95% CI 0.94 to 1.13) [2] comes from military populations, limiting generalizability to recreational runners [2].

Context-Specific Effects and Individual Differences

Different shoe features appear effective for different populations and injury types. The interaction between arch height and shoe type illustrates this principle: low-arched runners showed lower loading rates in motion control shoes, while high-arched runners benefited from cushion trainers [6]. Both findings may be correct within their respective margins, but a one-size-fits-all approach based solely on arch height oversimplifies the relationship.

Similarly, heel-to-toe drop effects depend on runner experience. Low drop shoes may benefit occasional runners but increase injury risk for regular runners [7]. This suggests dose-response or adaptation mechanisms where experienced runners have developed movement patterns and tissue adaptations suited to their habitual footwear, making transitions risky. The finding that medium-low drop reduces patellofemoral pain specifically ($p = 0.003$) [13] indicates that drop effects are injury-specific rather than universally beneficial.

The effect of midsole hardness varies by body mass [18], suggesting that optimal cushioning exists on a continuum rather than in discrete categories. Harder midsoles may reduce biomechanical risk factors for medial tibial stress syndrome, tibial stress fractures, Achilles tendinopathy, and iliotibial band syndrome [18], but this relationship likely depends on the runner's mass, strike pattern, and training load. Studies failing to account for these moderators may obscure real effects.

Biomechanical Mechanisms and Comfort Filter

The divergent findings may reflect that different shoe features operate through distinct mechanisms. Arch-based prescription assumes a biomechanical pathway where matching shoe support to arch height reduces excessive pronation and thereby injury risk. However, multiple high-quality studies found no reduction in injuries using this approach [1, 2, 12], suggesting either the theoretical mechanism is incorrect or the implementation (three broad categories) is too coarse to capture meaningful individual variation.

In contrast, shoe mass directly affects metabolic cost through mechanical work requirements, explaining the consistent 1% increase in oxygen consumption per 100g [4]. This relationship appears linear and independent of individual characteristics, making weight a more reliable optimization target.

Comfort may operate as an integrative filter. The finding that identical shoes received dramatically different ratings when labeled differently [8] demonstrates that psychological factors strongly influence perceived comfort. However, this doesn't negate comfort's potential value—runners may intuitively select shoes that minimize their individual injury risk, even if the mechanism isn't captured by traditional arch categorization. The lack of objective biomechanical differences despite comfort differences [8] suggests comfort integrates subtle factors beyond simple kinematic measures.

Sport-Specific Considerations

Basketball showed clear performance benefits from high traction ($p = 0.005$ for cutting performance) [5] and forefoot stiffness ($p = 0.013$ for sprinting) [5], but no effect of mass [5]. This contrasts sharply with running, where mass consistently affects economy [4]. The difference likely reflects the movement demands: basketball involves frequent changes of direction where traction dominates, while running emphasizes repetitive movements where even small mass penalties accumulate.

Field hockey players prioritized fit, comfort, support, and cushioning [11], with 63% believing stud design influences injury risk [11]. However, 46% of female players didn't use hockey-specific footwear [11], suggesting availability and fit challenges outweigh theoretical benefits of sport-specific design for some populations. This highlights that practical constraints (shoe availability, fit options) may limit the applicability of biomechanically optimal recommendations.

Reconciling Contradictions

The synthesis reveals that "shoe selection" is not a single question but multiple context-dependent questions:

For injury prevention in military training populations, arch-based prescription provides no benefit [1, 2]. The evidence here is strong (moderate to high certainty, large sample sizes, consistent findings across populations). However, for reducing specific injuries in specific populations—such as bone stress injuries in male military trainees—lighter shoes with lower drop and wider toe boxes reduce risk by 44% [3].

For performance optimization, minimizing mass while maintaining adequate comfort appears universally beneficial in running [4], while sport-specific features like traction dominate in basketball [5]. The strength of evidence varies by sport, with running having more extensive research.

For recreational runners, the lack of evidence for any particular selection paradigm [2, 4] combined with strong individual preferences [8] suggests prioritizing lightweight, comfortable shoes without assuming pronation control features are necessary [4]. This recommendation represents the best available synthesis given current evidence limitations.

The heterogeneity in findings reflects genuine differences in which shoe features matter for which outcomes in which populations, rather than contradictory results that need reconciliation through more research. Future studies should specify the target population, injury type, and biomechanical mechanism rather than seeking universal shoe selection rules.

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